Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/13/2023
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	Characterizing Plant Gravity Perception Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	Delaware	**State:**	OH
Zip Code:	43015-2333	**Congressional District:**	12
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	03/16/2015	**End Date:**	03/15/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 3/15/2022 per NSSC information (Ed., 3/16/21) NOTE: Extended to 3/15/2021 per NSSC information (Ed., 12/16/19) NOTE: Extended to 3/15/2020 per NSSC information (Ed., 12/17/18) NOTE: Extended to 3/15/2019 per NSSC information (Ed., 12/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AG55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plant growth is characterized by plasticity, allowing environmental cues like light and gravity to play a significant role in determining the overall form of the plant body. Statolith sedimentation in specialized cells represents one way that plants perceive gravity. Plants lacking functional statoliths show reduced but consistent gravity responses, suggesting the presence of a non-statolith sensory system. This proposal seeks to investigate the gravity perception machinery in roots of several genotypes of the higher plant Arabidopsis thaliana by taking advantage of the unique microgravity environment of Earth orbit on the International Space Station (ISS). One thrust of the work described here is to characterize the physiological properties of the putative non-statolith sensory system, including the threshold acceleration force required to activate it, and its capacity to modulate downstream growth regulation machinery. The second thrust is to determine whether the two gravity sensory systems control a single or multiple output systems. Addressing this question requires the microgravity environment in order to quantify the kinetics of gravitropism in roots having or lacking functional statoliths under fractional g conditions. The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions. In particular, the differences in growth responses and gene regulation between wild-type and starchless mutants will help elucidate the contributions of statolith-based sensing to plant gravity responses.
Task Progress:	Work in the final year of this task focused exclusively on analyzing the RNA-seq data generated from the frozen flight samples grown for 36 h under altered g conditions on the International Space Station (ISS) using the European Modular Cultivation System (EMCS) centrifuge. At this time, we have carried out exploratory data analysis in order to begin to understand the most fruitful approaches and comparisons to choose for the identification of differentially expressed genes and pathways. One key observation is the dramatic difference in the overall regulation of gene expression between wild-type (WT) and starchless (pgm-1) mutants. Plotting a heatmap of expression of the 1000 most variable genes reveals this difference in pattern, with WT seedling roots showing much more consistent regulation than starchless mutants.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Peer-reviewed Journals	Barker R, Kruse CPS, Johnson C, Saravia-Butler A, Fogle H, Chang HS, Trane RM, Kinscherf N, Villacampa A, Manzano A, Herranz R, Davin LB, Lewis NG, Perera I, Wolverton C, Gupta P, Jaiswal P, Reinsch SS, Wyatt S, Gilroy S. "Meta-analysis of the spaceflight and microgravity response of the Arabidopsis plant transcriptome." npj Microgravity. 2023 Mar 20;9(1):21. https://doi.org/10.1038/s41526-023-00247-6 ; PMID: 36941263; PMCID: PMC10027818 , Mar-2023
Articles in Peer-reviewed Journals	Canaday ELJ, Meyers A, Scinto-Madonich NJ, Wyatt SE, Wolverton C. "Physiological and transcriptomic assessment of Arabidopsis identifies two distinct gravity signaling systems." Plant Physiol. 2026 Feb 27;kiag095. Online ahead of print. https://doi.org/10.1093/plphys/kiag095 ; PMID: 41761687 , Feb-2026